KRISHNA AMANCHI

Mobile number: (+91) 9951629152.

krishna.a9171@gmail.com

Experience Summary:

- **Senior Software Engineer** with 8 years of experience in software development using C++98**, C+11**, and **C++14**.
- Experienced working with ARM and x86 controllers as well as Embedded Linux environments.
- Experienced in **multithreading** and **concurrent programming**.
- Skilled in **D-Bus**, Yocto build processes, and **CANoe** protocol.
- Proficient in using **CANoe** for simulation and **Yocto** for build automation.
- Developed and customized Linux-based embedded OS using the **Yocto Project**.
- Created custom **Yocto layers, recipes**, and **meta-data** for board bring-up and application integration.
- Experience in developing embedded software using C for a range of microcontrollers.
- Implemented TCP/UDP client-server communication systems.
- Experienced with **MQTT** protocol for IoT communication.
- Proficient in developing **Visual C++** applications in the Telecom domain.
- Experienced with creation **static** and **dynamic** libraries.
- Proficient in **CMake** build system and **Qt** Creator IDE.
- Good understanding of scripting languages such as **Python** and **Shell** scripting.
- Familiar with databases like **MySQL** and **SQLite**.
- Interfaced analog sensors with microcontrollers using **I2C/SPI** communication protocols.
- Proven skills in client interaction, project debugging, and onsite support.
- Experienced in supporting teams and ensuring consistent client service.
- Strong knowledge of communication and networking protocols.

Work Experience:

NeoTech Global,pvt Ltd -Hyderabad (Nov-2024 to till)

Project 1: TNG Simulation Framework for Spectrum Data Acquisition (Client: Thermo Fisher Scientific, Inc. Hyderabad.)

Role: Senior Software Engineer

D**escription:** Developed a simulation framework to emulate Thermo TNG hardware acquisition systems, enabling seamless testing of Tune applications without requiring physical instruments.

Key Responsibilities:

- Implemented firmware compatible with any native Linux OS.
- Developed an **SQLite** server application to record live data from the electrometer.
- Implemented **TCP/UDP** client-server communication for continuous spectrum streaming.
- Configured VirtualBox to run the simulated binary.

Skills:C++, Linux, Socket programming, CMake, Yocto, AWS, Lua,Git

Techmologics Innovation,pvt Ltd -Hyderabad (Dec-2022 to Oct-2024)

Project 2: CHARGING Systems (Client:Webasto Germany and Monrovia,US)

Role: Senior Software Engineer

Description: Worked on electric vehicle (EV) charging solutions for OEMs, focusing on intelligent charging stations and digital services to enable efficient and reliable EV charging.

Key Responsibilities:

- Developed, enhanced, and maintained automotive applications for EV charging systems.
- Utilized **CANoe** for network simulation and diagnostics.
- Created custom Linux builds using **Yocto Project**.
- Simulated real-time environments to support system analysis and improve product versions.
- Diagnosed and resolved field issues to ensure system stability and customer satisfaction.

Skills:C++, Linux, CAN, OCPP, CMake, Yocto, JIRA, Git,AWS, GoogleTest,CI/CD,QT

HCL Technologies Pvt Ltd -Chennai (Sep-2021 to Dec-2022)

Project 3: FIRE – Locomotive Control System (Client: Caterpillar)

Role: **Lead Engineer**

Description: Designed and implemented embedded software for real-time control and monitoring of locomotive operations. Integrated sensor systems for tracking critical parameters such as speed, temperature, and fuel efficiency.

Key Responsibilities:

- Developed, enhanced, and maintained locomotive control applications using **QNX RTOS**.
- Integrated various sensors and data acquisition systems for real-time parameter monitoring.
- Simulated real-time operating environments to support testing, analysis, and optimization.
- Performed debugging and resolved system-level issues to improve software reliability and performance.

Skills:C++, Linux, QNX, MQTT, JIRA, BitBucket, Git,CodeCollbarator,PRC,Source Tree.

Excel Technologies Systems Pvt. Ltd.-Hyderabad (FEB-2020 to May2021)

Project 4: Command Control Computer Communication Information (client RCI-DRDO Hyderabad)

Role: Software Developer

Description: Worked on the development of defence-grade distributed systems focused on real-time data processing and secure communication. The system processed complex geographical and graphical data in mission-critical environments.

Key Responsibilities:

- Designed, analysed, and developed applications for distributed defense systems.
- Implemented complex computational algorithms to process high-volume data, including geographical and graphical inputs.
- Utilized **Data Distribution Service (DDS)** protocol for reliable, real-time, publish-subscribe-based data communication across distributed components.

Skills:C++, Linux, DDS, QT, GIT.

Voice Gate Technologies India Pvt. Ltd.

Project 5: BATTERY MANAGMENT SYSTEM (OCT-2017 to Feb-2020)

Role:Software Programmer

Description: Measuring the voltage, current and temperature of consumer batteries of various ratings and there by transmitting this data wirelessly to server over RF.

Key Responsibilities:

- Designed and implemented a battery monitoring system to measure voltage, current, and temperature across various consumer battery configurations.
- Interfaced analog sensors with microcontrollers using **I2C/SPI** communication protocols.
- Developed firmware to collect and process real-time battery data.
- Integrated **RF modules** to enable wireless transmission of sensor data to a remote server for logging and analysis.

Skills:Embedded C, C++, GO, Windows, Linux, Sql, AWS, Git, MQTT, Vc++.

Technical Skills:

- Languages: **C++, Embedded C, Go lang.**
- Databases: MySQL, SQLite.
- Operating Systems: Windows, Linux. And macOS
- IDE Tools: Visual Studio, LiteX, Keil, Arduino, Qt, QNX Momentics, CLion, VSCode.
- Micro controllers: 8051, ATMega 328, ARM7, NodeMCU.
- Bus Protocols: UART, USART, SPI, I2C, RS485 and Canoe.
- Network Protocols: HTTP, HTTPS, FTP, MQTT, UDP, TCP/IP.
- Version Control Tools: Source Tree, Code Collaborator, Git,BIT Bucket.
- Project ManagmentTools: Azure, JIRA, PRC and Confluence.
- Debuggers: GDB, JTAG.
- Scripting languages: shell, Lua, Python.

Edcuation:

2012-2014 MSc (Electronics) from Osmania University with Aggregate of 80%.

Declaration:

I hereby declare that the information furnished above is true to the best of my knowledge

DATE: **(KRISHNA AMANCHI)**